

The Inversion Arc

Holographic Resolution in Asymmetric Coupled Systems

Constraint–Attractor Dynamics, Tensegrity, and the Banach–Tarski Connection

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Abstract

We develop the conceptual and structural consequences of the Asymmetric Codependent System (ACS) framework introduced in [1]. The *inversion arc* describes how a system that solves a constraint becomes the new constraint: the information asymmetry $\Delta\mathcal{I}$ flips sign as the system transitions from solution to dominant structure. We propose this as the *holographic resolution principle* and argue that it is instantiated by the Zamolodchikov *c*-theorem (2D QFT) and the Komargodski–Schwimmer *a*-theorem (4D QFT), which serve as proved examples of the inversion arc in quantum field theory. The *constraint–attractor cycle* connects the inversion arc to the gauge-algebraic results of [1]: the Palatini torsion-free condition $T^a = 0$ is both the attractor and the next constraint. Tensegrity structures provide a discrete geometric realisation of the ACS, where zero modes of the rigidity matrix correspond to gauge freedoms. The Banach–Tarski paradox is reinterpreted as a geometric ACS: the non-commuting rotations generate an exponential branching of states, and the emergent volume measures the 3rd-order holonomy.

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1 Introduction

This paper is the third in a trilogy. Paper A [1] develops the ACS framework and its application to gauge theory (with its accompanying verification suite). Paper B [2] applies the framework to number theory and the Riemann Hypothesis. This paper explores the universal structural consequence — the inversion arc — and requires only the BCH–TE morphism lemma of [1] and the definitions of Form, Function, and $\Delta\mathcal{I}$.

This paper develops the structural and conceptual consequences of the Asymmetric Codependent System (ACS) framework, whose mathematical foundations and primary physical application (the gauge-algebraic selection of $\mathfrak{su}(3)$ from Palatini geometry) are presented in [1]. The number-theoretic application is developed in [2].

The core observation is that in any system where two structurally different objects are codependent and asymmetric, the asymmetry *generates* observable pattern through the Lie bracket $[f, g]$ of the coupling operators. Here we explore what happens when that pattern becomes dominant: the solution becomes the constraint, and the information asymmetry inverts. This is the *inversion arc*.

1.1 Paper structure

Section 2 presents the tensegrity–gauge correspondence. Section 3 develops holographic resolution and the inversion arc. Section 3.3 presents the constraint–attractor cycle. Section 4 collects the algebraic foundations: the Killing-orthogonality theorem, the three-class spectral taxonomy of adjoint flows, and the algebraic non-traversability theorem (the ACS rereading of $ER=EPR$). Section 5 collects the unified claims across all ACS domains. Section B develops the Banach–Tarski connection.

Methodological note on epistemic compression

The results in this trilogy were obtained through repeated cycles of conjecture, explicit verification, and compression — a discipline formally articulated by Lakatos [10] as “proofs and refutations.” Several previously stated claims were narrowed during this process: a universal “ 2π inversion at three steps” was found to be representation-specific; the Wronskian on scalar zero modes was found to fail the Leibniz axiom (Paper B, the not-a-Poisson-bracket remark in its Section 2); a three-class rather than four-class spectral taxonomy emerges from Jordan–Chevalley decomposition (Section 4.2). Each correction tightens rather than weakens the framework. We distinguish throughout between proved theorems, conjectures explicitly flagged as open, and disproved hypotheses retained as boundary markers in the model space.

2 Discrete Geometry: Tensegrity as ACS

2.1 The tensegrity–gauge correspondence

A tensegrity network has nodes $\{q_i\} \subset \mathbb{R}^n$, edges partitioned into cables (tension) and struts (compression), with energy:

$$H = \sum_{(i,j) \in \text{cables}} k_{ij} V_+(|q_i - q_j|, L_{ij}) + \sum_{(i,j) \in \text{struts}} k_{ij} V_- (|q_i - q_j|, L_{ij}), \quad (1)$$

where V_+ penalises extension and V_- penalises compression.

Lemma 2.1 (Zero modes are gauge freedoms). *The zero eigenvalues of the rigidity matrix [5] $K = RR^\top$ (where $R_{3i,e}$ is the unit vector of edge e at node i) correspond to deformations that cost zero energy. In the ACS correspondence, these align with the gauge redundancies of the lattice gauge theory defined on the same graph: directions in configuration space where $\Delta\mathcal{I} = 0$.*

Numerical verification: the icosahedral tensegrity (12 bulk nodes, 6 struts) yields exactly 6 zero modes, matching the $\dim(SO(3)) + \dim(\text{translations}) = 6$ gauge redundancies of a 3D lattice gauge theory.

The ACS structure in this case:

- Form = cable network (tension, pulls toward equilibrium: structural).
- Function = strut network (compression, maintains separation: dynamic).
- Coupling = node positions q_i shared by both.
- Asymmetry: $V_+(r, L) \neq V_-(r, L)$ — the cable and strut potentials are structurally different nonlinear functions.
- Emergent pattern: the equilibrium configuration, which is neither the minimiser of cables alone nor struts alone.

3 Holographic Resolution

3.1 The inversion arc

Definition 3.1 (Holographic Resolution). Let C be a constraint and S a system that solves C by reducing $\Delta\mathcal{I}$. S achieves *holographic resolution* of C if:

- (HR-1) **Boundary completeness:** $I(\partial S; C) = H(C)$ — the mutual information between the boundary ∂S and the constraint equals the full entropy of C ;
- (HR-2) **Interior redundancy:** $I(\partial S; \text{int}(S) \mid C) = 0$ — given C , the boundary carries no information about the interior beyond what C already determines;
- (HR-3) **Outward gating:** $\Delta\mathcal{I}(S \rightarrow \text{exterior}) < 0$ — the boundary gates information outward, not inward.

The Ryu–Takayanagi formula $S_{\text{EE}} = \text{Area}(\partial A)/(4G)$ is recovered when $\Delta\mathcal{I} = 0$: at information balance, the entanglement entropy of a bulk region equals the area of its boundary in Planck units. Condition (HR-1) is the information-theoretic restatement of the RT formula; (HR-2) is the “no-hair” property of the holographic encoding; (HR-3) is the statement that holographic screens gate access from inside to outside, not the reverse.

Definition 3.2 (Self-Resolving Structure). A structure S is *self-resolving* if it contains an intrinsic measure $\mu(t)$ such that whenever S begins to invert its information asymmetry ($\Delta\mathcal{I} \rightarrow -\Delta\mathcal{I}$), the value $\mu(t)$ becomes nonzero and is legible *within* S ’s own record — not only detectable by external observers.

Remark 3.3. The Bianchi identity $D_{[\mu}F_{\nu\rho]} = 0$ is the prototype self-resolving structure: any deviation from gauge-covariant curvature is immediately encoded in the field equations themselves. A magnetic monopole would violate $\nabla \cdot B = 0$, and this violation is legible in the same layer as the field — not detectable only post hoc.

3.2 The inversion theorem

Theorem 3.4 (Inversion arc of access systems). *Let S be a system solving constraint C_0 by achieving $\Delta\mathcal{I} > 0$ (Form drives Function: open, outward-flowing). As S becomes the*

dominant access structure, new actors enter the field only through S . At this point $S = C_1$ is the new constraint, and:

$$\Delta\mathcal{I}(S \text{ as solution}) > 0 \quad \longrightarrow \quad \Delta\mathcal{I}(S \text{ as constraint}) < 0. \quad (2)$$

The information asymmetry has inverted. The key has become the lock.

Proof. The proof follows from the BCH–TE morphism lemma of [1].

Step 1 (Role assignment). When S is a solution, let $f = S$ (the outward-flowing operator, Form) and $g = C_0$ (the constraining operator, Function). By hypothesis, $\Delta\mathcal{I}(f \rightarrow g) > 0$: Form drives Function.

Step 2 (Role reversal). When S becomes the dominant access structure, the roles swap: $f' = C'_0$ (the new dynamic actors, now Function) and $g' = S$ (now the constraining boundary, Form). The operators are the same but their positions in the ACS coupling are exchanged: $f' \leftrightarrow g'$.

Step 3 (BCH sign reversal). The BCH–TE morphism gives $\Delta\mathcal{I}(f, g) = \varepsilon \mathbb{E}[f - g] + \mathcal{O}(\varepsilon^2)$. Under role reversal $f \leftrightarrow g$:

$$\Delta\mathcal{I}(g, f) = \varepsilon \mathbb{E}[g - f] = -\varepsilon \mathbb{E}[f - g] = -\Delta\mathcal{I}(f, g) + \mathcal{O}(\varepsilon^2).$$

The first-order transfer entropy flips sign exactly. The second-order term $2\varepsilon^2 \langle [f, g], \cdot \rangle$ is likewise odd under the exchange, since $[g, f] = -[f, g]$; remaining corrections are $\mathcal{O}(\varepsilon^3)$ and do not alter the sign for sufficiently small coupling.

Step 4 (Conclusion). The sign of $\Delta\mathcal{I}$ inverts when S transitions from solution to constraint. The inversion rate τ^{-1} scales with the adoption rate (how quickly S becomes the dominant access layer). $\square$

Remark 3.5 (Computational verification). Three instances of the inversion arc are verified in [1]:

1. The Ricci flow uniformisation ($41\times$ reduction in the curvature standard deviation): the Ricci flow (solution) becomes the Einstein equation (constraint).
2. The constraint hierarchy $\|G\| > \|D\| > \|H\|$ on the spin network: the Gauss constraint (Form) gates the diffeomorphism and Hamiltonian constraints (Function).
3. The Wheeler–DeWitt attractor ($\ker(\hat{H})$ weight 50.7%): the quantum state flows to $\hat{H}\Psi = 0$, where the WdW equation (solution) becomes the definition of the physical Hilbert space (constraint).

Historical instances (Table 1):

Domain	ACS Form	ACS Function
2D QFT (c-theorem)	UV fixed point	IR RG flow
4D QFT (a-theorem)	UV couplings	Weyl anomaly
Gauge theory	vierbein e	connection ω
Quantum gravity	triad $\tilde{E}$	Ashtekar A
Number theory	primes $\{p_n\}$	Riemann zeros $\{\gamma_k\}$

Table 1: ACS instances across domains. The RG examples (top two rows) provide established instances of monotone information flow (Zamolodchikov c-theorem [3] and the Komargodski–Schwimmer a-theorem [4]), anchoring the framework in known results.

3.3 The constraint–attractor cycle

The inversion arc (Theorem 3.4) has a dynamical interpretation that unifies the attractor structure of the gauge theory, $\mathrm{GL}(4)$, and quantum gravity sections of [1].

In the Palatini formulation, the field equations drive the system to $T^a = 0$ (torsion-free geometry). This equilibrium is both the *attractor* of the ACS flow and the *constraint* on subsequent evolution: once torsion vanishes, the connection is fully determined by the vierbein, and the system has no remaining degrees of freedom at 1st order.

Breaking this constraint — activating $T^a \neq 0$ — is the ACS analog of Form overcoming Function. The excess torsion produces chiral zero-modes ([1]), which carry the spinor structure needed for complexification. Those spinor degrees of freedom self-organise into the next constraint: the colour gauge symmetry $\mathrm{SU}(3)$, whose confinement mechanism provides the new attractor.

The cycle is:

$$\begin{array}{l} T^a = 0 \xrightarrow{\text{torsion activates}} T^a \neq 0 \xrightarrow{\text{chiral modes}} \text{spinor bundle,} \\ \text{spinor bundle} \xrightarrow{\text{complexify}} \mathfrak{su}(3) \xrightarrow{\text{confine}} \text{new constraint.} \end{array}$$

Each stage is one step of the inversion arc. The Lie bracket $[f, g]$ at each stage measures the asymmetry driving the transition; when it vanishes, the system is at the attractor; when it is non-zero, the system is being driven toward the next attractor.

For quantum gravity, the same structure appears: the Wheeler–DeWitt condition $\hat{H}|\Psi\rangle = 0$ (the WdW conjecture [1]) is the quantum attractor where $\Delta\mathcal{I}_Q = 0$. Breaking this balance ($\hbar$ corrections, $\Delta\mathcal{I}_Q \neq 0$) produces the arrow of time — the quantum version of activating torsion.

For the Riemann Hypothesis, the attractor is the critical line $\sigma = \frac{1}{2}$: the state where the prime-zero ACS is stationary (the spectral stationarity theorem [2]). Off-critical zeros correspond to the system not having reached equilibrium — the variance grows (Section 5), the attractor has not been found.

In every domain, the constraint is the attractor, and the attractor is the next constraint. The ACS framework makes this cycle computable: $\Delta\mathcal{I}$ measures how far the system is from equilibrium, and the Lie bracket $[f, g]$ measures what is driving it there.

4 Algebraic Foundations of the Inversion Arc

The inversion arc, as developed in Section 3, has the structural form of a Lie bracket whose successive applications saturate at finite order. This section makes that statement precise. We prove the Killing-orthogonality theorem (Section 4.1), which is the algebraic content of “non-traversability” in the ACS framework. We classify adjoint flows by spectral type (Section 4.2). We re-read the ER=EPR conjecture in this language (Section 4.3). Together these results sharpen what the inversion arc *is* mathematically, and what it is not.

4.1 The Killing-orthogonality theorem

Theorem 4.1 (Killing-orthogonality). *For any matrix Lie algebra $\mathfrak{g}$ and any $X, Y \in \mathfrak{g}$,*

$$\mathrm{tr}([X, Y] X) = 0, \quad \mathrm{tr}([X, Y] Y) = 0. \quad (3)$$

Equivalently, the bracket output $B = [X, Y]$ is trace-orthogonal to both of its inputs in the sense of the Killing form.

Proof. By trace cyclicity, $\text{tr}(XYX) = \text{tr}(YX^2)$, so $\text{tr}([X, Y]X) = \text{tr}(XYX) - \text{tr}(YX^2) = 0$. The second identity follows by symmetric exchange. $\square$

This identity is verified symbolically in `src/paper_c/killing_orthogonality.py` (returning 0 in SymPy) and numerically across 1,000 random pairs in each of $\mathfrak{sl}(3, \mathbb{R})$ and $\mathfrak{sl}(4, \mathbb{R})$ with maximum residual $\sim 10^{-15}$.

Chirality hopping example

In $\mathfrak{sl}(4, \mathbb{R})$ with the standard basis $\{H_i, A_{ij}, S_{ij}\}$ where H_i are Cartan generators, A_{ij} are antisymmetric, and S_{ij} are symmetric off-diagonal, the bracket

$$[H_1, A_{01}] = 2S_{01} \quad (4)$$

maps a Cartan element and an antisymmetric off-diagonal element to a symmetric off-diagonal element — crossing chirality sectors exactly. This is a numerical verification (exact integer match) of the more general fact that the bracket of sym and anti sectors lands in the opposite-chirality sector. The Killing-orthogonality theorem then forces the result S_{01} to be Killing-orthogonal to both H_1 and A_{01} , which is verified directly: $\text{tr}(S_{01}H_1) = \text{tr}(S_{01}A_{01}) = 0$.

The three ambiguity numbers in $\mathfrak{sl}(4, \mathbb{R})$

Given $B = [X, Y]$, the inverse problem “recover (X, Y) from B ” has three distinct ambiguity numbers depending on what one means by recovery.

- (a) **Passive ambiguity (subspace).** The Killing-orthogonality theorem forces both X and Y to lie in the codimension-1 subspace $B^\perp = \{Z \in \mathfrak{sl}(4) : \text{tr}(BZ) = 0\}$. Its dimension is $\dim B^\perp = n^2 - 2 = 14$.
- (b) **Generic active ambiguity (Jacobian kernel).** The bracket map $\mu : \mathfrak{sl}(4) \oplus \mathfrak{sl}(4) \rightarrow \mathfrak{sl}(4)$, $\mu(X, Y) = [X, Y]$ has differential $d\mu(\delta X, \delta Y) = [\delta X, Y] + [X, \delta Y]$. At a generic point (X, Y) , $\text{rank } d\mu = n^2 - 1 = 15$ and $\dim \ker d\mu = 15$ — the locally surjective regime. Verified across 20 random pairs in `src/paper_c/orthogonal_complement_probe.py`.
- (c) **Degenerate active ambiguity at (H_1, A_{01}) .** At the specific Cartan-antisymmetric pair, where the bracket lands in a 1-dimensional subspace, the differential degenerates to $\text{rank } d\mu = 11$ and $\dim \ker d\mu = 19$.

These three numbers measure different things and should not be conflated. Citing only one “ambiguity factor” without context produces misleading statements.

4.2 The three-class spectral taxonomy

The orbits of $\exp(t \text{ad}_X)$ on $\mathfrak{g}$ classify by the spectrum of ad_X via the Jordan–Chevalley decomposition $X = X_s + X_n$ with X_s semisimple, X_n nilpotent, and $[X_s, X_n] = 0$. The semisimple part X_s further splits over $\mathbb{R}$ into eigenvalues that are either real-valued or complex-conjugate pairs, yielding three distinct flow types.

Theorem 4.2 (Spectral taxonomy of adjoint flows). *Any finite-dimensional real linear ACS operator X admits a Jordan–Chevalley decomposition into commuting semisimple and nilpotent parts. The semisimple part further partitions adjoint flows into three classes:*

- (i) **Elliptic** — ad_{X_s} has imaginary eigenvalues (paired conjugates). $\exp(t \text{ad}_X)$ is rotational and periodic.
- (ii) **Hyperbolic** — ad_{X_s} has real eigenvalues. $\exp(t \text{ad}_X)$ is exponential growth/decay; no 2π loop closure.

(iii) **Parabolic** — $X_s = 0$, $X_n \neq 0$. $\exp(t \operatorname{ad}_X)$ is polynomial.

The generic case is a commuting sum of these classes acting on distinct invariant subspaces.

The taxonomy is mathematically standard [6, 7]; its relevance here is to clarify that the three-fold closure $\operatorname{ad}^3 = (16/9) \operatorname{ad}$ in $\mathfrak{sl}(4, \mathbb{R})$ (the three-fold Cayley–Hamilton saturation established in the torsion-hierarchy appendix of Paper A [1]) is an algebraic statement about the minimal polynomial of $\operatorname{ad}_{T_{B-L}}$ — not a geometric “ 2π inversion at three steps.” The latter applies only in the elliptic case.

ACS instance	Spectrum of ad_X	Class	Geometric content
SU(2) quaternion	$\{\pm i\}$	Elliptic	2π rotation = $-I$
Frenet–Serret operator	$\{0, \pm i\sqrt{\kappa^2 + \chi^2}\}$	Elliptic	curvature/torsion rotation
$\mathfrak{sl}(4, \mathbb{R})$ Palatini (T_{B-L})	$\{0, \pm 4/3\}$	Hyperbolic	boost-like, no 2π closure
Core rope ring ($R^3 = R$)	$\{-1, 0, +1\}$	Hyperbolic	$\mathbb{Z}/3$ permutation

Table 2: Four ACS instances classified by spectral type. Elliptic systems exhibit literal 2π inversion under three-step composition; hyperbolic systems exhibit only algebraic Cayley–Hamilton saturation, with no rotational closure. The Frenet–Serret operator A satisfies $A^3 = -(\kappa^2 + \chi^2)A$, an elliptic-class saturation; the Palatini adjoint satisfies $\operatorname{ad}_{T_{B-L}}^3 = (16/9)\operatorname{ad}_{T_{B-L}}$, a hyperbolic-class saturation. Each identity is verified in `src/paper_c/spectral_taxonomy.py`.

Remark 4.3 (On the universal- 2π overclaim). An earlier draft of this work proposed that “three-step bracket chains universally produce 2π geometric inversions” across all ACS systems. Verification against $\operatorname{ad}_{T_{B-L}}$ in $\mathfrak{sl}(4, \mathbb{R})$ shows otherwise: $\exp(2\pi \operatorname{ad}_{T_{B-L}})$ has spectral norm of order $e^{8\pi/3} \approx 4,348$, not unity. The flow does not close. What is universal is finite-order Cayley–Hamilton saturation; the specific geometric content (rotation versus boost versus shear) depends on the spectral class. The corrected statement is preserved here as Theorem 4.2.

Remark 4.4 (Grading selection from spectral class). The spectral taxonomy has a direct consequence for $\mathbb{Z}/2$ gradings: in the companion note [11], it is shown that minimising a universal class of adjoint spectral functionals over symmetric involutions on $\mathbb{R}^n$ selects the grading aligned with the minority eigenvalue cluster of the Cartan generator. For compact algebras (elliptic class), the minimiser is the trivial grading; for noncompact split algebras (hyperbolic class), it is a nontrivial $(k, n-k)$ grading. The three-class taxonomy of Theorem 4.2 thus determines not only the dynamics (rotational, exponential, polynomial) but also the preferred grading structure compatible with each spectral class. (See [11] for the Coleman–Mandula constraint on interpreting this grading as physical spacetime signature.)

4.3 Algebraic non-traversability: the ACS rereading of ER = EPR

The Maldacena–Susskind ER = EPR conjecture [8] posits that entangled quantum systems are connected by non-traversable Einstein–Rosen bridges. In the conjecture’s original formulation, non-traversability is imposed as an external causality constraint: local operators on either side commute (preserving relativistic causality across the bulk geometry).

The ACS framework derives the operational content of non-traversability from the bracket algebra alone. No bulk geometry is invoked.

Proposition 4.5 (Algebraic non-traversability). *Let $B = [X, Y]$ be the bracket output of an ACS coupling. The projection $\pi_X : \mathfrak{g} \rightarrow \operatorname{span}\{X\}$ defined by $\pi_X(M) = \frac{\operatorname{tr}(MX)}{\operatorname{tr}(X^2)} X$*

satisfies $\pi_X(B) = 0$ identically. No information from B can be “traversed” back to X through scalar inner-product reconstruction.

Proof. By Theorem 4.1, $\text{tr}(BX) = 0$. Hence the projection coefficient is zero. $\square$

This is verified at machine precision in `src/paper_c/er_epr_algebraic.py` (projection coefficient $|c| < 10^{-16}$ for random (X, Y) pairs in $\mathfrak{sl}(4, \mathbb{R})$). Combined with the standard quantum-mechanical fact that local Alice and Bob operators on a Bell state commute (verified in the same module), this gives the operational content of non-traversability without invoking bulk geometry: the ACS bracket structure is intrinsically “opaque” from inputs to outputs in the trace inner product.

Remark 4.6 (Limits of the rereading). This algebraic non-traversability is a statement about the bracket algebra, not a derivation of holographic entropy bounds or the Ryu–Takayanagi formula [9]. Whether the algebraic non-traversability of ACS bracket outputs can be quantitatively matched to AdS/CFT entanglement-wedge depth saturation is an open question requiring construction of an explicit correspondence map. We flag this as a research direction, not a derived consequence.

5 Unified Claims and Open Problems

5.1 The unified object

All six tabulated domains (gauge theory, gravity, number theory, discrete geometry, renormalisation group, and quantum gravity) are projections of a single structure: an ACS on a space where Form and Function are the two independent fields, with the bracket $[f, g]$ generating curvature and the holonomy $[[f, g], \cdot]$ being the emergent pattern.

Domain	Form	Function ω	Bracket $[f, g]$	Pattern
Gauge theory	$F_{\mu\nu}$	A_μ	$[A_\mu, A_\nu]$	YM equation
Gravity	$e^a{}_\mu$	$\omega^{ab}{}_\mu$	$R = d\omega + \omega^2$	Einstein eqs.
Number theory	$\{p_n\}$	$\{\gamma_k\}$	Explicit formula	RH stationarity
Tensegrity	Cables	Struts	Equilibrium	Zero modes
RG flow	UV couplings	IR couplings	β -function	c -theorem
Quantum gravity	$\tilde{E}_i^a$	A_a^i	LQG constraints	Spin foam

Table 3: All six domains as projections of the ACS framework.

5.2 Open problems

Conjecture 5.1 (Standard Model from $\text{GL}(4)$ fiber). Let $\mathcal{Y}^{14}$ be the bundle of positive-definite metrics over M^4 , with fiber $\text{GL}(4)/O(4)$ (dimension 10) and base M^4 (dimension 4). The Lie algebra $\mathfrak{gl}(4)$ acting on each fiber satisfies:

$$\mathfrak{gl}(4) \supset \mathfrak{so}(3, 1) \supset \mathfrak{su}(2) \oplus \mathfrak{u}(1).$$

Conjecture: with the ACS coupling $(\mathbf{F}, \Phi) = (e, \omega)$ on $\mathcal{Y}^{14}$, the asymmetry bracket $[f, g]$ generates $\mathfrak{su}(2) \oplus \mathfrak{u}(1)$ as the electroweak sub-algebra of the ACS asymmetry (see [1] for the proof that $\mathfrak{su}(3)$ cannot arise from the $O(4)$ fiber alone).

Conjecture 5.2 (Geometric fermions from torsion). In an ACS with $(e^a{}_\mu, \omega^{ab}{}_\mu)$ and non-zero torsion $T^a \neq 0$, the torsion–spin coupling produces modes with half-integer angular momentum from purely geometric data. No independent spin structure is required; the Cartan–Kibble–Sciama mechanism realises fermions as the 1st-order ACS coupling between Form and Function.

Theorem (Zamolodchikov c -theorem [3], restated in ACS language). For any unitary 2D QFT, there exists a function $C(g)$ on coupling space such that $dC/dt \leq 0$ along RG flow, with equality if and only if the theory is at a fixed point. At fixed points, C equals the Virasoro central charge c . In the ACS framework, fixed points correspond to $\Delta\mathcal{I} = 0$ (information balance between UV Form and IR Function). The 4D analogue (Komargodski–Schwimmer a -theorem [4]) extends this monotonicity to four-dimensional QFT.

5.3 Gaps revealed by formalisation (updated status)

Several gaps identified in the original draft have been addressed, with varying levels of rigour. We distinguish three epistemic tiers: *theorem* (rigorous proof from stated axioms), *derived* (conditional result with numerical verification and mechanism identified), and *conjecture* (matching pattern without full derivation).

- (1) **The SM generation conjecture: conjecture.** The companion paper [1] identifies one full SM generation from the $GL(4)$ fiber: all 8 electric charges and 8 hypercharges match, including a right-handed neutrino. Three generations follow from Jacobi truncation of the BCH at order 3. The Koide angle $\theta_0 = \arctan(\lambda_{\text{Wolfenstein}})$ to 0.23%, linking the lepton mass hierarchy to the Cabibbo angle. *Not yet: a field-theoretic derivation showing that the BCH orders map to distinct fermion families in the quantum theory.*
- (2) **The Wheeler–DeWitt attractor: derived.** A 3-node spin network Lindblad steady state concentrates in $\ker(\hat{H})$ with $\langle \hat{H} \rangle = 3 \times 10^{-6}$. This is a numerical result on a toy model; the full spin-foam limit is open.
- (3) **The Higgs potential shape: derived.** The $\Delta\mathcal{I}$ landscape across the $GL(4)$ fiber produces the Mexican Hat potential for 15.9% of Palatini directions. The quartic coupling $\lambda = 2\sqrt{3}/27$ matches the SM value to 0.85%. *Not yet: a derivation of the electroweak scale v from first principles.*
- (4) **The neutrino see-saw: conjecture.** Three layers of suppression produce $m_\nu \times M_R = m_e^4/(9m_\tau^2)$ (holds to 0.1%), predicting $M_R \sim 49$ keV. The Dodelson–Widrow mechanism overproduces this state by $\sim 10^5$, requiring either subdominant DM or an alternative production mechanism.
- (5) **Definition 3.1 and “emergent”:** formalised with mutual information bounds (HR-1 through HR-3).
- (6) **Theorem 3.4 (testability):** the inversion arc now has three verified physical instances (Ricci flow, spin network constraint hierarchy, and the Wheeler–DeWitt attractor).
- (7) **Torsion coupling hierarchy: theorem.** All 15 generators of $\mathfrak{sl}(4, \mathbb{R})$ classified into Tier 0 (zero coupling, 9 generators) and Tier 2 (coupling $32/9$, 6 generators). Read on the electroweak-adapted generators J_i, K_i — which are equal-weight mixtures of a Tier-0 and a Tier-2 generator, and therefore carry $\frac{1}{4} \cdot \frac{32}{9} = \frac{8}{9}$ — the unbroken, electroweak and colour-lepton sectors stand in the ratio 0:1:4. This is a statement about that generator set, not a third eigenvalue of $\text{ad}_{T_{B-L}}$, whose spectrum is two-valued (Paper A, torsion-hierarchy section [1]).

- (8) **Bosonic vacuum energy cancellation: theorem.** $\sum_{X \in \mathfrak{sl}(4)} \|[T_{B-L}, X]\|^2 \cdot K(X, X) = 0$ exactly, in rational arithmetic. Reduces the cosmological constant problem from 10^{121} to $\sim 10^{55}$ (Paper A, torsion-hierarchy appendix [1]).
- (9) **Corrections from the exploration session (updated).** (a) $\tan \beta = 1/2$ was a misidentification of the Yukawa ratio $\tilde{h}/h \approx 2/3$ with the VEV ratio. (b) The W/Z torsion share is a conditional prediction, not a theorem. (c) $\lambda = 2\sqrt{3}/27$ is an algebraic result (Koide projection), not an RG fixed point. (d) The α_2 cross-coupling $\text{Tr}(\Phi^\dagger T^A \Phi) \text{Tr}(\Delta_R^\dagger T^A \Delta_R)$ vanishes identically in the minimal bi-doublet because $T^A \Phi = 0$ for $\Phi \sim (1, 2, 2)$ (the bi-doublet is an $\text{SU}(4)$ singlet); α_2 is therefore forbidden by representation theory, not eliminated by choice. (e) The β_c term is excluded at tree level in the minimal model: extremization forces $\tan \beta = \pm 1$, which by the no-go identity $M_u - M_d = (h - \tilde{h})(\kappa_1 - \kappa_2)$ gives $V_{\text{CKM}} = I$, contradicting observed quark mixing. Either $\beta_c = 0$ at tree level or the Higgs sector must be extended (Branch B with $\Sigma \sim (15, 1, 1)$).
- (10) **The irreducible boundary: 6 inputs in Branch A (minimal bi-doublet).** The bracket algebra and consistency filters together fix the Pati–Salam Higgs sector to:
- Locked by Palatini: $\lambda_\Phi = 2\sqrt{3}/27$, $2\rho_1 + \rho_2 = 16/9$, $g = 4/3$, $\tilde{h}/h = 2/3$, $N_{\text{gen}} = 3$.
 - Forbidden by representation theory: $\alpha_2 = 0$.
 - Excluded at tree level: $\beta_c = 0$ (else $V_{\text{CKM}} = I$).
 - Free at tree level: ρ_1, α_1 .
 - Radiatively generated: $\tan \beta$ (value pending Coleman–Weinberg).
 - Calibrations: v, v_R .
- Total inputs: 4 free + 2 calibrations = 6 (vs. 19+ in the SM, a reduction factor of $\sim 3.2\times$). Whether one-loop Coleman–Weinberg corrections fix $\tan \beta$ uniquely (reducing the count to 5) is open.
- (11) **Three-layer self-pruning of the Higgs sector.** The reduction from 5 quartic couplings to 2 free + 1 radiative proceeds through three independent consistency filters: representation theory ($\alpha_2 = 0$), vacuum extremization (β_c forces $\tan \beta = \pm 1$ if present), and flavor phenomenology ($\kappa_1 = \kappa_2 \Rightarrow M_u = M_d$). Each operates at a different mathematical layer and constrains the others; this is not a coincidence but a structural feature of the minimal bi-doublet embedding.

6 The Inversion Arc as a Universal Principle

The constraint–attractor cycle

$$\text{Constraint} \rightarrow \text{Solution} \rightarrow \text{Dominance} \rightarrow \text{Inversion} \rightarrow \text{New Constraint}$$

is instantiated across every domain touched by the ACS framework. Table 4 maps each domain to its specific realisation of the cycle, with the computational verification from [1] indicated where available.

The universality of this cycle is not merely an analogy but a structural consequence of the BCH–TE morphism: any system with codependent Form and Function and non-zero bracket $[f, g] \neq 0$ will exhibit sign reversal of $\Delta\mathcal{I}$ under role exchange (Theorem 3.4). The cycle terminates at $\Delta\mathcal{I} = 0$ attractors: confinement in gauge theory, $\ker(\hat{H})$ in quantum gravity, stationarity in number theory, the VEV in the Higgs sector.

A Historical Precursors: Newton, Newlands, and Mersenne

Domain	Constraint	Solution	Inversion	New constraint
Gauge theory	$T^a = 0$ (torsion-free)	Activate torsion	Chiral modes	$\mathfrak{su}(3)$ confinement
Quantum gravity	$\hat{H}\Psi = 0$ (WdW)	$\hbar$ corrections	Lindblad flow	Physical Hilbert space
Geometry	Curved manifold	Ricci flow	$41 \times (\text{std})$	Einstein equation
Riemann zeros	$\sigma = \frac{1}{2}$ (RH)	Off-critical zeros	$\mathcal{S}(N, X) \rightarrow \infty$	Stationarity restored
Higgs sector	$V'(\phi) = 0$	Radial perturbation	Sombrero trough	$\text{VEV} = v$

Table 4: The inversion arc across domains. Each row instantiates the same cycle: a constraint generates a solution; the solution becomes dominant; dominance inverts the information asymmetry; the inverted system becomes a new constraint. The first three rows have computational verification in [1].

The ACS framework has unexpected antecedents in three historical observations used as interpretive comparisons; they do not establish physical identity.

A.1 Optical color and internal color: corrected distinction

Optical color mixing and internal $\text{SU}(3)$ charge refer to different state spaces. The three fundamental weights form a triangle in a Cartan weight diagram, but the balanced superposition of their basis states remains a triplet. Zero Cartan expectation does not make it an invariant vector. The singlet occurs in appropriate tensor products, not as the sum of the three fundamental basis states. The earlier optical-wheel identification and the inference of confinement from it are withdrawn. A visual analogy may motivate a calculation; it cannot replace that representation test.

A.2 The element spiral and atomic octaves (de Chancourtois, Newlands)

Two documented pre-Mendeleev arrangements anticipate the structure used here: in 1862 A.-E. Béguyer de Chancourtois proposed a helical (“telluric screw”) arrangement of the elements, and in 1865 John Newlands observed the *law of octaves* — chemical periodicity with an octave structure, the noble-gas positions later emerging as the “silent” nodes. (An earlier draft attributed this to “Alexander Reina Russell, 1867”; we could not document such a figure and have replaced the attribution with the standard history.)

In the ACS framework:

- Shell capacities $2n^2$ (2, 8, 18, 32) arise from $\text{SO}(3)$ representation theory: $\sum_{l=0}^{n-1} (2l+1) = n^2$, doubled for spin;
- Noble gases ($\Delta\mathcal{I} = 0$) are the ACS *attractors* — closed-shell configurations where the electron Form and nuclear Function are in perfect information balance;
- Chemical reactivity ($\Delta\mathcal{I} \neq 0$) is the departure from the attractor;
- The element spiral is read as the ACS layered resolution: each “octave” is one BCH order of the atomic Hamiltonian, with the noble gas at its $\Delta\mathcal{I} = 0$ fixed point.

A.3 Dimensional devolution

The ACS framework inverts the traditional micro-to-macro construction. The fundamental object is the full $\mathfrak{sl}(4)$ fiber (15 dimensions); particles are *projections* onto our 3+1D observation space. The compression sequence

$$\mathfrak{sl}(4) [15\text{D}] \rightarrow \mathfrak{su}(3) [8\text{D}] \rightarrow \text{Cartan} [2\text{D}]$$

contains a computationally checked projection, not a singlet construction: the Cartan projection kills all off-diagonal structure exactly ($\max |M_{ij}| = 0.000000$ for $i \neq j$). A

perfectly ordered geodesic on T^{15} has autocorrelation 0.95 in 15D but only 0.23 when projected to 3D. This example demonstrates loss of a chosen correlation under projection; it does not derive quantum randomness.

B The Banach–Tarski Paradox as a Geometric ACS

The Banach–Tarski paradox is the ultimate mathematical manifestation of a non-commutative system generating emergent volume. One takes a solid ball in $\mathbb{R}^3$, partitions it into five pieces, and reassembles them using rotations and translations to obtain two identical copies of the original ball. The mechanism relies entirely on the fact that the rotation group $SO(3)$ contains a free subgroup on two generators a and b . These generators do not commute: $ab \neq ba$. The set of all words in a and b forms an infinite binary tree; each distinct sequence gives a distinct orientation. The exponential growth of words (2^n words of length n) is what allows a single piece to be “unrolled” to cover the whole ball.

Mapping this to the ACS:

- **Form** and **Function** are the two rotation generators a and b .
- The **2nd-order bracket** $[a, b] \neq 0$ is exactly the non-commutativity that creates the free group.
- The **3rd-order holonomy** is the irreducible residue of the closed loop $aba^{-1}b^{-1}$ etc. — the fact that no finite combination of generators returns to the identity unless it is trivial. This is the source of the paradoxical decomposition.
- The **emergent volume** is the extra measure (the second ball) generated by the non-commutative operations. In ACS language, the information asymmetry $\Delta\mathcal{I}$ measures precisely this geometric defect: the failure of the flow to close under composition of the two generators.

Thus Banach–Tarski serves as a concrete mathematical illustration of the ACS mechanism, where the fields are rotation operators and the measure is the Lebesgue measure on the ball. The paradox shows that when $[f, g] \neq 0$, the system can create new phase-space volume, and $\Delta\mathcal{I}$ provides a quantitative measure of this non-closure across the other domains considered here.

Amendment: scope and evidence, 24 September 2026

Killing-form identities and algebraic projections retain their stated mathematical scope. A Cartan projection is not a singlet projection, and optical color addition does not construct an invariant vector of the fundamental $SU(3)$ representation. The optical identification is withdrawn below. Physical holography, confinement and particle identification require a state space and dynamics in addition to these algebraic maps. Reduced scalar-potential formulas apply only to their restricted ansatz: the complete invariant audit has 17 quartics, hidden transverse directions and nontrivial running.

For the complete-action and microscopic-carrier audit, see the companion papers *Complete Scalar Actions and Identifiability Boundaries in the ACS Framework* and *Condensate Binding, Charge Protection, and the Boundary of Particle Identification* in this repository’s `papers/updates/` directory. The dated publication record separates fresh portable checks from preserved archival results; it does not claim a complete rerun of every historical experiment or a proof of RH.

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